Solar System Fact Sheet

A concise reference of physical and orbital facts about the Sun and the eight planets. Figures are rounded, widely published values drawn from public-domain astronomical data. Distances from the Sun are averages; day lengths are given in Earth hours and orbital periods in Earth days or years.

The Sun

The Sun is the star at the center of the Solar System. It is not a planet. It holds about 99.86 percent of the total mass of the Solar System. Its diameter is roughly 1,391,000 kilometers, about 109 times the diameter of Earth. The temperature at its visible surface, the photosphere, is close to 5,500 degrees Celsius, while its core reaches about 15 million degrees Celsius. The Sun is composed mainly of hydrogen and helium, and it generates energy by nuclear fusion of hydrogen into helium.

Mercury

Mercury is the smallest planet and the closest to the Sun. Its diameter is 4,879 kilometers. It has no moons and no substantial atmosphere. A single orbit around the Sun takes 88 Earth days, the shortest year of any planet. Because it rotates slowly, one solar day on Mercury lasts about 1,408 hours. Surface temperatures swing from about 430 degrees Celsius in daylight to about minus 180 degrees Celsius at night.

Venus

Venus is the hottest planet, with a mean surface temperature near 465 degrees Celsius, hot enough to melt lead. This heat is caused by a dense atmosphere of carbon dioxide that traps energy through a runaway greenhouse effect. Venus has a diameter of 12,104 kilometers, close to that of Earth, and is sometimes called Earth's twin. It has no moons. Unusually, Venus rotates in the opposite direction to most planets, a motion called retrograde rotation.

Earth

Earth is the third planet from the Sun and the only body known to support life. Its diameter is 12,756 kilometers. It completes one orbit in 365.25 days and has a single natural satellite, the Moon. About 71 percent of its surface is covered by water. Its atmosphere is roughly 78 percent nitrogen and 21 percent oxygen. Surface gravity is 9.8 meters per second squared.

Mars

Mars is known as the Red Planet because iron oxide, or rust, gives its surface a reddish color. Its diameter is 6,779 kilometers, about half that of Earth. Mars has two small moons, Phobos and Deimos. A day on Mars lasts 24.6 hours, only slightly longer than an Earth day, while its year is 687 Earth days. Surface gravity is 3.7 meters per second squared, about 38 percent of Earth's. Mars is home to Olympus Mons, the tallest volcano in the Solar System, rising about 22 kilometers above the surrounding plains.

Jupiter

Jupiter is the largest planet, with a diameter of 139,820 kilometers, more than eleven times that of Earth. It is a gas giant composed mainly of hydrogen and helium, with no solid surface. Jupiter has dozens of moons; the four largest, Io, Europa, Ganymede, and Callisto, were discovered by Galileo in 1610 and are called the Galilean moons. A prominent feature is the Great Red Spot, a giant storm that has raged for at least several centuries.

Saturn

Saturn is famous for its bright, extensive ring system, made mostly of ice and rock particles. It is the second largest planet, with a diameter of 116,460 kilometers. Saturn is the least dense planet; its average density is lower than that of water, so it would float if placed in a large enough ocean. Like Jupiter, it is a gas giant and has many moons, the largest of which, Titan, is bigger than the planet Mercury.

Uranus

Uranus is an ice giant with a diameter of 50,724 kilometers. Its most distinctive property is that it rotates on its side, with an axial tilt of about 98 degrees, so that its poles take turns facing the Sun over its long orbit. Uranus appears blue-green because methane in its atmosphere absorbs red light. It takes 84 Earth years to orbit the Sun.

Neptune

Neptune is the farthest planet from the Sun, at an average distance of about 4.5 billion kilometers. Its diameter is 49,244 kilometers. Neptune has the strongest winds in the Solar System, reaching about 2,100 kilometers per hour. One orbit takes about 165 Earth years. Neptune was the first planet located by mathematical prediction rather than direct observation, in 1846.

A Note on Pluto

Pluto was considered the ninth planet from its discovery in 1930 until 2006, when the International Astronomical Union reclassified it as a dwarf planet. It orbits in the Kuiper Belt, a region of icy bodies beyond Neptune.